

Yibo Yin

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EDUCATION	École polytechnique fédérale de Lausanne (EPFL) <i>M.Sc. in Digital Humanities, School of Computer and Communication Sciences (IC)</i> • GPA: 5.56 /6.00 Lausanne, Switzerland 09/2025 - Present
	Wuhan University <i>B.Eng. in Computer Science and Technology</i> • GPA: 3.93 /4.00 (top 4%) Wuhan, China 09/2021 - 06/2025
PUBLICATIONS	MATStruct: High-Quality Medial Mesh Computation via Structure-aware Variational Optimization Ningna Wang, Rui Xu, Yibo Yin , Zichun Zhong, Taku Komura, Wenping Wang, Xiaohu Guo <i>ACM SIGGRAPH Asia 2025 (Conference Track)</i>
RESEARCH EXPERIENCE	Image and Visual Representation Lab, EPFL <i>Research Assistant PI: Prof. Sabine Süsstrunk ; Supervisor: Zhuoqian Yang, Doctoral Assistant </i> 06/2026 - Present • Investigating weight space generation for implicit neural representations, with the goal of learning modality-agnostic priors over neural fields across multiple data modalities.
	Computer Graphics Lab, The University of Texas at Dallas <i>Research Assistant (Remote) Advisors: Prof. Xiaohu Guo  and Dr. Ningna Wang </i> 04/2024 - 11/2025 • Adapted Loop subdivision surface evaluation and sharp feature fitting algorithms to structured medial mesh. • Designed a simplification algorithm for structured medial mesh to preserve geometric structure under subdivision. • Proposed a subdivision surface fitting method for structured medial mesh, targeting sharp-feature preservation during fitting.
	Graphics and Vision Lab, Wuhan University <i>Research Assistant Advisor: Prof. Chunxia Xiao </i> 05/2023 - 01/2024 • Built an RGB-ToF/depth data acquisition pipeline and collected 10,000+ paired image-sensor samples. • Proposed a novel monocular depth estimation method using RGB images and auxiliary ToF/depth signals.
	Physically Based Ray Tracer <i>CS-440 Advanced Computer Graphics Rendering Competition, EPFL</i> 2026 • Extended the Nori framework with heterogeneous volumetric path tracing, delta and ratio tracking, Henyey-Greenstein phase functions, and MIS over emitter, phase, and BSDF sampling. • Implemented HDR environment map lighting, thin-lens depth of field, motion blur, principled BSDF, and GGX rough dielectric materials.
SELECTED PROJECTS	Differentiable Rendering Project <i>Realistic Graphics Lab, EPFL Mentor: Dr. Ziyi Zhang </i> 03/2026 - 06/2026 • Developed a differentiable Gaussian Process Implicit Surface (GPIS) sampling module in Dr.Jit to support inverse-rendering with implicit geometry. • Developed a hybrid surface-volume rendering approach for Many-Worlds inverse rendering.
AWARDS	• Outstanding Student (10% school-wide), Wuhan University 2022, 2023, 2024 • Second Class Scholarship (10% school-wide), Wuhan University 2022 • Third Class Scholarship (15% school-wide), Wuhan University 2023, 2024 • Lei Jun (Xiaomi) Computer Innovation and Development Fund , Wuhan University 2024
SKILLS	• Languages: Mandarin (Native), English (TOEFL iBT 102) • Programming Languages: C++, C, Python, C#, GLSL, SQL, Verilog HDL, Java, $\text{\LaTeX}$ • Libraries / Frameworks / Tools: PyTorch, Dr.Jit, Mitsuba, PBRT-v3, Blender, Eigen, CGAL, libigl, Git, CMake